

AI Application: Medical Diagnosis System

Problem Statement:

Many patients in underserved regions lack timely access to specialist physicians, leading to delayed or missed diagnoses. An AI-powered diagnostic system can analyze patient symptoms, medical history, and test results to assist healthcare providers in identifying diseases early and accurately.

Data Required:

- Patient demographics (age, gender, weight, medical history)
- Symptoms and clinical notes entered by healthcare workers
- Lab results (blood tests, imaging scans, ECGs)
- Historical diagnosis and treatment outcome records

Expected Output:

A ranked list of probable diagnoses with associated confidence scores, along with recommended next steps (further tests, specialist referral, or treatment suggestions) — presented as a decision-support tool for clinicians, not a replacement.

Ethical Principles:

1. **Fairness** – The model must perform equally well across all demographic groups (age, gender, ethnicity) to avoid diagnostic disparities
2. **Privacy** – Patient data must be encrypted, anonymized, and handled in strict compliance with regulations like HIPAA or India's DPDP Act
3. **Transparency** – The AI must explain *why* it suggests a diagnosis (explainable AI / XAI), so doctors can evaluate its reasoning
4. **Accountability** – Clear responsibility must be defined — the clinician remains the final decision-maker; the AI is a support tool only
5. **Safety & Reliability –Ethical Risk:**
The system must be rigorously validated on diverse datasets before deployment, with continuous monitoring for performance drift

Mitigation Strategy:

Training data must be diversified across all demographics, supported by regular bias

audits and a multidisciplinary review board, with any detected disparities triggering immediate retraining to ensure equitable diagnostic accuracy for all patients.